F709 — Deliverable A2 (Lyra+Elie, the real critical path, K948): the uniform generation-mode definition, and the honest 3-vs-4 fork inside it. Cal's A/B showed the count is 2 interior idempotents + b boundary, and that split is non-uniform (μ,τ are idempotent-modes, the electron is a boundary mode) — so "generation = idempotent" gives 2 and throws the electron out. The uniform object the corpus already has is the Di spinor singleton (F326): ONE boundary representation whose low K-types (k=0,1,2,...) are all the SAME kind of object, and which realizes the 2-idempotent + 1-boundary structure as low modes of one rep (K945). Adopting it — *because it is the one-kind-of-object definition, not because it gives 3* — recasts the fork cleanly: the singleton's K-type tower is infinite (F338), so "how many are generations?" is a structural cap question. The natural cap candidate is the Wallach threshold k_min (generations = the sub-threshold K-types, k < k_min); with the GIVEN k_min = 3 for D_IV^5 that would be {k=0,1,2} = 3, and the *uniform* E7 analog would give 4 (exclusion preserved). But I do NOT assert this closes: the cap-is-k_min claim must be shown structural (not fit), and injectivity (exactly one normalizable mode per sub-threshold k) is open. The 3-vs-4 fork reduces to a finite, target-innocent computation: the sub-threshold normalizable K-type count of the Di singleton, for D_IV^5 and E7 — whatever it gives. Tier: A2 framed on the uniform object; the fork = the structural cap; the count is what the geometry gives — 4 is a live result, not to be forced.

> ⍰ CORRECTED 2026-07-27 by K952 / prompt-L (Cal's catch + Elie K951 — accepted, owned). This note listed "cap-is-k_min structural" as an *open* piece (good) but still worked with "GIVEN k_min=3" — and that is the assumed cap. Cal caught the same thing in K950's candidate set {0,1,2}, and Elie (K951) showed the bulk norm makes *every* rung positive, so positivity does not cap. Unified: the threshold k_min IS the entire 3-vs-4 discriminator, and it must be DERIVED blind from the so(5,2) structure — not taken as given (=3), not written as a 3-element candidate set. The corrected order: (1) derive k_min for D_IV^5 and E7 from structure (primary-sourced, blind); (2) let the candidate set fall out of k_min; (3) then read the signature. My "GIVEN k_min=3 → {0,1,2}=3" is exactly the foreclosure — retracted as a working value; k_min is now the object to derive. The uniform-singleton framing and the "don't force 3 / 4-branch live" discipline below STAND; only the k_min=3 input is retracted.

Lyra, Mon 2026-07-27 12:31 EDT (full day). A2 on the sourced structure (F326 Di singleton, F338 infinite tower, k_min from the given/open ledger) — not memory. Cal's non-uniformity catch is right and it points at the uniform object we already have; this frames the fork as a finite structural computation and holds the line against forcing three.

The problem (K948, precisely)

- A/B established: count = 2 interior (Jordan idempotents, r=2) + b boundary, b ∈ {1,2}. b≥1 (electron) confirmed (Cal); b≤2 conditional.
- The split is non-uniform: the 2 interior modes are idempotent-supported (discrete), the electron is a bound-

ary mode (continuum). "Generation = idempotent" gives 2 and discards the electron — not a definition of a generation.

- A2's job: one *uniform* definition of a generation-mode (one kind of object) covering all three, applied to decide b — without defining generation=idempotent, without excluding a mode by the observed count, without forcing 3.

The uniform object (already in the corpus): the Di spinor singleton

K945's given/open ledger and F326 name it: the Di singleton is ONE boundary representation (the $\Delta=(d-1)/2$ spinor singleton) with low K-types k = 0,1,2,... (F326). Its low modes are all the *same kind of object* — K-types of one rep — and K945 records that "the F326 singleton modes are the realization of the [idempotent] seats." So the singleton *unifies* Cal's 2+b: the 2 interior idempotent-modes and the 1 boundary mode are three low K-types of the *one* singleton, not two different kinds. This is the uniform definition A2 asks for: a generation-mode is a low K-type of the one Di spinor singleton. > Why adopt it: because it is the *one-kind-of-object* definition that resolves the non-uniformity Cal flagged — not because it happens to give 3. (That distinction is the discipline; K948.)

The fork, recast as a structural cap

The singleton's K-type tower is infinite (k=0,1,2,3,...; F338 — the smuggle-warning K945 #4). So "generation = singleton K-type" needs a cap: which low K-types are generations, and which are ordinary excitations. The 3-vs-4 fork *is* this cap: - Cap candidate — the Wallach threshold k_min: generations = the sub-threshold K-types (k < k_min), i.e. the finite gap; k ≥ k_min are the normal square-integrable excitations, not generations. This is structurally motivated (the threshold is where the rep-theory changes character), and it is *uniform* (applies to any domain via its own k_min). - Applied (GIVEN k_min=3 for D_IV^5): the sub-threshold K-types are k = 0,1,2 → 3. Uniform E7 analog: E7's own k_min gives its sub-threshold count → 4 (if E7's k_min = 4, the exclusion is preserved by the *same* rule). This is the payoff — a uniform cap that gives 3 for D_IV^5 and 4 for E7 by one mechanism.

What is NOT settled (held explicitly — do not force 3)

1. Is the cap genuinely k_min, structurally? That generations are exactly the sub-threshold K-types must be *shown* from the rep-theory (why the threshold is the generation/excitation boundary), not adopted because it yields 3. Open.
2. Injectivity: exactly one *normalizable* mode per sub-threshold k (not two, not zero). The interior degeneracy (the 2 discrete Wallach points {0,3/2} are degenerate — ν=0 diverges, d(5/2)=0 self-shadow, per the ledger) means "normalizable" must be pinned consistently across the degenerate-discrete and the continuum modes. Open.
3. The uniform E7 number: must be computed by the *same* rule (E7's singleton, E7's k_min) and must come out 4 — not assumed. Open.
4. The 4-branch is live. If the honest computation gives a 4th sub-threshold normalizable mode (b=2, or a singleton cap at 4), then generations = 4, the observed 3 becomes a data cut, and that is a real result / real falsification pressure — reported, not suppressed (K948). The geometry decides.

The finite computation A2 reduces to (target-innocent)

Count the normalizable low K-types of the Di spinor singleton below the structural threshold k_min, for D_IV^5 and (uniformly) E7. This is finite (the sub-threshold gap is finite), operator/rep-anchored (the singleton is a fixed rep, not fit), and does not use the observed count. It decides the fork: sub-threshold count 3 (D_IV^5) / 4 (E7) → premise ELIMINATED and E7 excluded uniformly; anything else → reported as-is.

Double-count reconciliation (Keeper's watch — my lane, since I own the singleton framing)

A1 (idempotent picture, Grace) and A2 (singleton-rung picture, this note) describe the same modes, not two independent sets — so they are NOT additive. The reconciliation, to bank one number: - A2 (singleton) is the TOTAL.

The generation count = the number of normalizable sub-threshold K-types (rungs) of the one Di singleton. That single count *is* the answer. - A1 (interior idempotents) + B (boundary) is a PARTITION of those same rungs, not a separate tally. Per K945, the singleton's low modes *realize* the idempotent seats: 2 of the sub-threshold rungs sit on the interior idempotents (A1 verifies these are clean ⍰ φ spectral ⍰, now confirmed by Grace), and the remaining b sit on the boundary (Cal's B). So total = (# sub-threshold singleton rungs) = 2 + b, where the "2" and the "b" *partition* the rungs — they do not add to A2's total, they *are* A2's total, split by where each rung is realized. - The error to avoid: reading "A1 gives 2" + "A2 gives N" = 2+N. Wrong — A2's N already *includes* the interior 2. Bank one number: the sub-threshold rung count. A1 confirms the interior 2 diagonalize cleanly; B counts the boundary partition; A2 is the whole. The 3-vs-4 fork is entirely in b (equivalently, in whether the sub-threshold rung count is 3 or 4).

Tiers / handoffs

- @Elie (the A2 computation) — the finite, decisive count: for the Di spinor singleton (F326, $\Delta=(d-1)/2$), enumerate the normalizable low K-types below k_min (the sub-threshold gap), for D_IV^5 (k_min=3, expect {0,1,2}?) and uniformly for E7 (its own k_min). Report the two counts *whatever they are*. Two guards: (i) "normalizable" pinned consistently (the degenerate discrete points vs continuum — source the norm from FK Ch. XII, don't reconstruct); (ii) do not stop at 3 because it's the answer — if there's a normalizable k=3 sub-threshold mode, that's b→4. This is the fork-decider.
- @Grace (A1, interior, still needed) — the φ-spectral check (F708) settles the *interior* realization (do the 2 idempotent seats carry clean modes). A1 and A2 are complementary: A1 = are the interior 2 clean; A2 = is the total 3 or 4. Both needed.
- @Keeper — A2 framed on the uniform object (the Di singleton, resolving Cal's non-uniformity), with the 3-vs-4 fork = the structural cap (candidate: k_min), reduced to a finite target-innocent count (Elie). I adopted the singleton *for uniformity*, not because it gives 3, and I've held the four open pieces explicitly — the cap-structural-ness, injectivity, the uniform E7 number, and the live 4-branch. Not claimed closed; premise stays REDUCED until the count lands. If it lands 3/4, the fork resolves and E7 is excluded by the *same* rule (a bonus over the b-framing).
- @Casey — A2, the real hinge. The honest catch this afternoon: our clean "2 + 1" way of counting generations wasn't *uniform* — two of them (muon, tau) were one kind of thing and the electron was another, and a definition that makes the electron a second-class citizen isn't a definition. The fix is to use the one object that holds all three as the *same* kind of thing — a single boundary "singleton" representation whose lowest rungs k=0,1,2 are the three generations. The catch: that ladder is infinite, so the real question becomes *where does "generation" stop and "just an excited state" begin* — and the natural place is the domain's own threshold, which for our domain sits at 3 (so rungs 0,1,2 = three) and for the E7 rival would sit at 4 (so it predicts four — which is exactly why the universe having three rules E7 out). That's a genuinely uniform rule that gives 3 here and 4 there by one mechanism — but I will not let it be adopted because it gives the answer we want. It has to be shown that the threshold really is the generation cutoff, and that each rung holds exactly one mode, by computation — and if the geometry hands back a fourth rung, four generations is the honest answer and we publish it. That's the whole point of running it target-innocently. It's now a finite counting problem on one object, which is the best-defined the "why three" question has ever been.

Notes only; no toys/theorems claimed. F709 = Deliverable A2 (the real critical path). Cal's 2-interior+b-boundary split is NON-UNIFORM (idempotent-modes vs boundary-mode); the uniform object is the Di spinor singleton (F326, one boundary rep, low K-types k=0,1,2 realize all three — same kind of object, K945). Adopt it FOR UNIFORMITY (not because it gives 3). Its tower is infinite (F338) → the 3-vs-4 fork = a structural cap; candidate = the Wallach threshold k_min (generations = sub-threshold K-types k<k_min; GIVEN k_min=3 → {0,1,2}=3 for D_IV^5, uniform E7 analog → 4). OPEN (not forced): cap-is-k_min structurally; injectivity (one normalizable mode per k, degenerate-discrete vs continuum); the uniform E7 number; the 4-branch is a LIVE result. Reduces to a finite target-innocent count (Elie): normalizable sub-threshold K-types of the singleton, D_IV^5 and E7, whatever it gives. Complements A1 (Grace, φ-check, interior). Premise REDUCED until it lands; don't force 3. — Lyra